

PI-LSTM + DACP

Physics-Informed Deep Learning with
Dual Adaptive Conformal Prediction

Streamflow Prediction & Uncertainty Quantification
Across 5 Snowmelt-Driven Watersheds

Presentation Overview

01	Problem Statement	Why streamflow prediction matters and the challenge
02	Literature Review	Key prior work informing our approach
03	Study Sites & Data	5 watersheds, 1,461 daily observations each
04	PI-LSTM Architecture	Model design, physics loss, attention mechanism
05	Training & Feature Engineering	Walk-forward, Optuna, site-specific parameters
06	Model Performance	R^2 , RMSE, MRE, NSE — what each tells us per site
07	DACP Method	Dual adaptive regulators for prediction intervals
08	Coverage Experiments	70%, 80%, 90%, 95% across all sites
09	Per-Site Analysis	Plots, metrics, and discussion for each watershed
10	Conclusion & Deliverables	Summary and what we produced

SECTION 1

Problem Statement

Why Streamflow Prediction Matters

The Problem

Snowmelt-driven watersheds provide critical water supply for agriculture, municipalities, and ecosystems across the Western U.S.

Accurate daily streamflow forecasting is essential for:

Water resource management — reservoir operations, irrigation scheduling, and drought planning.

Flood warning — peak discharge events can cause devastating damage if not anticipated.

Ecological monitoring — aquatic ecosystems depend on minimum flow thresholds.

The Challenge

Snowmelt hydrology is highly nonlinear. Discharge depends on SWE, temperature, precipitation history, and soil moisture.

Traditional models struggle because:

Time-dependent interactions — snowpack accumulates over weeks but can melt in days.

Regime shifts — the same watershed behaves differently during snowmelt season vs. rain-on-snow vs. baseflow.

Our approach: Physics-Informed LSTM for point prediction + DACP for distribution-free uncertainty quantification.

SECTION 2

Literature Review

Literature: Snowmelt & Streamflow

Moghaddasi et al. (2025)

Snow drought → hydrologic drought progression using ML. SWE during winter highly impacts spring/summer streamflow. Optimal lookback period: 60–90 days. Shows that snowpack is a leading indicator of drought.

Relevance: Confirms SWE as primary driver — our WESD and SNWD features capture this relationship.

Kiewiet et al. (2022)

Spatial/temporal variability of rainfall + snowmelt on streamflow in rain–snow transition zones. Strong spatial correlation ($r^2 = 0.88$) between modeled and observed surface water input. Annual streamflow mainly controlled by precipitation timing.

Relevance: Supports our use of precipitation features (prcp_7d, API) alongside snow data.

Literature: PIML & DACP

Maharjan et al. (2025)

Physics-informed ML for snowpack decline prediction. Compared SWAT, LSTM, and PIML — found PIML provides the most reliable performance. Used PIML with CMIP6 climate projections for future snowpack scenarios.

Relevance: Validates PI-LSTM as the architecture of choice for snow-driven hydrology.

Fu / Zong et al. (2026)

Dual Adaptive Conformal Prediction (DACP) for uncertainty quantification. Introduces two adaptive regulators — a data partitioning regulator and a quantile regulator — that handle non-exchangeable data and distribution shifts. Distribution-free: no parametric assumptions required.

Relevance: The DACP method we apply — developed by our research group.

SECTION 3

Study Sites & Data

Five Snowmelt-Dominated Watersheds

All sites: 1,461 daily observations (2019–2022), Western U.S.

Bunchgrass Meadow

High-flow, snowmelt-dominated

0 – 4,500 cfs

Extreme dynamic range; largest discharge variance. Required $\log(1+Q)$ transform.

Easy Pass

High-flow, glacial/snow

0 – 275 cfs

High-elevation alpine; rapid solar-driven melt onset in spring. GHI is key driver.

Paradise

Mid-flow, rain-on-snow

0 – 35 cfs

Flashy rain-on-snow spikes; hardest timing to predict.

Touchet

Low-flow, rain-dominant

0 – 680 cfs

Rare extreme peak events amidst stable low baseflow.

Swift Creek

Low-flow, complex

0 – 170 cfs

Lowest predictability; complex mixed hydrology. No physics term.

SECTION 4

PI-LSTM Model Architecture

Model Architecture: PI-LSTM

Architecture

Two Stacked LSTM Layers

128 hidden units per layer. Captures temporal dependencies over a 14-day input window.

Multi-Head Attention (4 heads)

Dynamically weights the most critical time steps — rapid temperature spikes or rain-on-snow events.

Fully Connected Layers

128 → 64 → 32 with ReLU, dropout 0.2. Outputs single scalar discharge prediction.

Physics-Informed Loss

$$L_{\text{total}} = L_{\text{data}} + \lambda \cdot L_{\text{physics}}$$

Physics term penalizes violations of daily water balance. λ is site-specific (set to 0 where physics adds noise).

Input Features (8 total)

WESD	Water Equivalent Snow Depth
GHI	Global Horizontal Irradiance
PRCP	Daily Precipitation
SNWD	Snow Depth
TAVG	Average Temperature
prcp_7d / API	7-day Precip or Antecedent Index
doy_sin	Day-of-year sine (seasonality)
doy_cos	Day-of-year cosine (seasonality)

Input Features: What Goes Into the Model?

Each site uses 8 input features in a 14-day sliding window. Four key engineered features explained below.



prcp_7d (7-Day Cumulative Precipitation)

Sum of the last 7 days of precipitation (rain + snowmelt). Captures how much water entered the watershed recently. A rising prcp_7d signals an incoming discharge spike — soil is saturating.

Used by: Easy Pass, Paradise, Touchet (F1 feature set)



API (Antecedent Precipitation Index)

Exponentially-weighted sum of past rainfall — recent days count more, older days decay. Represents how 'wet' the soil already is. High API = saturated ground = even small rain triggers big runoff.

Used by: Bunchgrass Meadow, Swift Creek (F2 feature set)



sin(day of year) — Seasonal Sine

Converts day-of-year (1–365) into $\sin(2\pi \times \text{day}/365)$. Peaks in spring (~day 91), lowest in fall (~day 274). Tells the model 'where in the year' without a harsh Jan→Dec jump.

Captures seasonal snowmelt timing — spring thaw, summer recession



cos(day of year) — Seasonal Cosine

The cosine companion: $\cos(2\pi \times \text{day}/365)$. Peaks in winter (~day 1), lowest in summer (~day 182). Together with sine, encodes a full circular calendar — all four seasons, no ambiguity.

Sin + cos together = complete seasonal encoding

Training: Walk-Forward Prediction

Walk-Forward (Expanding Window)

Model is trained on data up to day t , predicts day $t+1$, then expands the training window and retrains. This prevents temporal data leakage — the model never sees future data. It simulates real-world forecasting conditions.

Key Parameters

Input window:	14 days
LSTM layers:	2×128
Attention:	4 heads
Train / Test:	85% / 15%
Optimizer:	Adam
Max epochs:	300
Patience:	30
Dropout:	0.2

Reproducibility & Optimization

Fixed random seed (`torch.manual_seed(42)`) before each model creation ensures reproducibility across retraining passes.

Scalers refit on current data at each retrain step to prevent data leakage — scaler never sees future data.

Validation split: 10% during walk-forward retraining (reduced from 15% to preserve training samples on smaller windows).

Optuna hyperparameter tuning: 30–50 trials per site to select loss function, learning rate, and architecture details.

Site-Specific Configuration

Each site was individually configured based on Optuna trials and hydrological characteristics:

Site	Feature Set	Loss Function	λ (Physics)	Transform	Why This Configuration
Bunchgrass	F2 (API k=0.92)	NSE	0 (disabled)	log(1+Q)	Widest range; NSE normalizes by variance. Log stabilizes training. Physics disabled in log space.
Easy Pass	F1 (prcp_7d)	Huber $\delta=1.0$	Site-specific	None	Alpine melt driven by GHI. Huber limits peak underprediction influence on training.
Paradise	F1 (prcp_7d)	Huber $\delta=1.0$	Site-specific	log(1+Q)	Rain-on-snow spikes. 7-day precip captures rain patterns. Huber robust to outlier peaks.
Touchet	F1 (prcp_7d)	Huber $\delta=1.0$	Site-specific	None	One massive peak (~680 cfs). Huber prevents peak from dominating training.
Swift Creek	F2 (API k=0.92)	MSE	0 (disabled)	None	Complex hydrology. Water balance hurt low-flow performance. Plain LSTM works better.

F1 (prcp_7d): 7-day cumulative precipitation — rain-responsive sites. **F2 (API):** Antecedent Precipitation Index (k=0.92) — snowmelt-dominated sites with longer moisture memory.

SECTION 5

Model Performance Results

PI-LSTM Performance Summary

Site	Feature Set	Loss	RMSE (cfs)	MRE	R ²	NSE
Bunchgrass Meadow	F2 (API)	NSE	238.73	0.380	0.790	0.790
Easy Pass	F1 (prcp_7d)	Huber	15.73	0.492	0.545	0.545
Paradise	F1 (prcp_7d)	Huber	4.31	0.886	0.677	0.677
Touchet	F1 (prcp_7d)	Huber	11.00	0.505	0.893	0.893
Swift Creek	F2 (API)	MSE	5.93	0.259	0.276	0.276

14-day rolling input, walk-forward prediction, site-specific feature sets and loss functions. No lagged discharge feature.

Best Performer: Touchet

R² = 0.893 — captures 89% of daily variance. Huber loss was critical to prevent the single extreme peak from dominating training.

Most Challenging: Swift Creek

R² = 0.276 — complex mixed hydrology. Water balance constraint hurt performance. Uses plain LSTM ($\lambda = 0$). But best MRE (0.259).

Understanding RMSE & MRE

RMSE (Root Mean Squared Error)

Average error magnitude in the same units as discharge (cfs). **Scale-dependent** — you must compare to the site's discharge range.

Bunchgrass's 238.73 cfs looks huge — but on a site ranging 0–4,500 cfs, it's only ~5% of max.

Swift Creek's 5.93 cfs seems small — but on a site that only reaches ~170 cfs, it's ~3.5% of max.

RMSE alone cannot compare sites — you need scale context.

MRE (Mean Relative Error)

Normalizes errors by observed value — **scale-independent**. MRE = 0.380 means predictions are off by ~38% on average.

Swift Creek has best MRE (0.259) despite worst R^2 — errors are small in absolute terms relative to observed values.

Paradise has worst MRE (0.886) — flashy rain-on-snow spikes cause large relative errors even when absolute errors are small.

Understanding R^2 / NSE

R^2 / NSE (Variance Explained)

Fraction of observed discharge variability explained by the model. $R^2 = 1.0$ is perfect prediction.

Touchet $R^2 = 0.893$ — model captures ~89% of daily discharge variance. Strong baseline prediction.

Swift Creek $R^2 = 0.276$ — only ~28% explained. Complex mixed hydrology resists LSTM-based prediction.

$NSE = R^2$ here because we use the same formulation ($1 - MSE/Var$).

Key Takeaway: No Single Metric Tells the Whole Story

Touchet has best R^2 (0.893) but moderate RMSE (11.00 cfs).

Swift Creek has worst R^2 (0.276) but best MRE (0.259) — errors are small relative to observed values.

Bunchgrass has highest RMSE (238.73 cfs) but strong R^2 (0.790) — large errors on a very large scale site.

Always interpret metrics together in the context of each site's discharge scale and variability.

SECTION 6

Dual Adaptive Conformal Prediction (DACP)

Why Uncertainty Quantification?

Point Predictions Are Not Enough

A prediction of "tomorrow's flow = 45 cfs" gives no indication of confidence.

Is it 45 ± 2 cfs? Or 45 ± 50 cfs?

Decision-makers — flood managers, reservoir operators, ecologists — need to know how much to trust each prediction. Without uncertainty bounds, they cannot properly assess risk.

This is especially critical during regime transitions (snowmelt onset, rain-on-snow events) when prediction errors are largest.

DACP wraps any point-prediction model with adaptive prediction intervals.

Distribution-free — no Gaussian assumptions. Adaptive — intervals respond to model accuracy. Coverage guarantees — e.g., 95% of true values fall within bounds.

DACP: Building Prediction Intervals

At each time step, DACP uses past prediction errors to build an interval around the new prediction:

Step 1: Collect nonconformity scores

Compute $|\text{actual} - \text{predicted}|$ for the past 180 days (rolling window). This gives a fixed-size set of recent errors.

Step 2: Partition errors via the data partitioning regulator t_m

$(1 - t_m)$ fraction of errors goes to calibration, the rest to a reference set.

Step 3: Compute the interval half-width

Take the $(1 - \alpha_m)$ -quantile from the calibration set. This quantile q becomes the half-width.

Step 4: Prediction interval

$[\max(\hat{y} - q, 0), \hat{y} + q]$ (floor at 0 because discharge cannot be negative)

DACP: Dual Adaptive Regulators

After observing the true value, DACP updates both regulators:

Data Partitioning Regulator (t_m)

If covered: $t_{m+1} = t_m + \gamma \cdot \mu_1$ ($\mu_1 = 0.03$, slow adjustment)

If missed: $t_{m+1} = t_m + \gamma \cdot \mu_2$ ($\mu_2 = 0.50$, fast reaction)

Controls how much of the error history goes to calibration vs. reference.

Quantile Regulator (α_m)

$$\alpha_{m+1} = \alpha_m + \lambda(\alpha^* - e_m)$$

where e_m is the error indicator (1 if missed, 0 if covered). $\lambda = 0.01$ ensures very slow quantile updates to prevent oscillation.

$\mu_2/\mu_1 \approx 17\times$ — the algorithm reacts fast to misses but slowly when covered. This asymmetry is key.

DACP Parameter Choices

Parameter	Value	Why This Value
Rolling window	180 days	~117 calibration points — stable quantile estimation with temporal relevance
t_m (initial)	0.35	65% of error history goes to calibration — stable quantiles
t bounds	[0.10, 0.60]	Keeps calibration fraction between 40–90%
μ_1 (covered)	0.03	Slow: ~33 covered points to meaningfully change t_m
μ_2 (missed)	0.50	Fast: ~2 missed points trigger significant widening
k	1 – target	0.05 for 95%, 0.30 for 70% target
λ	0.01	Slow quantile update — prevents oscillation
Calibration	Rolling 180-day	Most recent (1–t_m) fraction of 180-day window

Final DACP: Fixed Parameters Across All Sites

Site	n	t _m init	t bounds	μ_1	μ_2	k_adap	λ	Coverage	Avg Width
Bunchgrass	180	0.35	[0.10, 0.60]	0.03	0.50	0.05	0.01	94.6%	816.4 cfs
Easy Pass	180	0.35	[0.10, 0.60]	0.03	0.50	0.05	0.01	94.5%	47.0 cfs
Paradise	180	0.35	[0.10, 0.60]	0.03	0.50	0.05	0.01	94.6%	14.9cfs
Touchet	180	0.35	[0.10, 0.60]	0.03	0.50	0.05	0.01	94.4%	28.3 cfs
Swift Creek	180	0.35	[0.10, 0.60]	0.03	0.50	0.05	0.01	93.8%	9.8cfs

Evaluation Metrics for Prediction Intervals

Coverage Rate

Expected coverage — fraction of actual values inside the interval.

Target: match the nominal level (e.g., 95%). Below target = intervals too narrow. Above target = intervals wider than necessary.

Average Width

Average prediction interval width in cfs. Lower is better IF coverage is met.

Trivially wide intervals give 100% coverage but are useless. Must be interpreted relative to each site's discharge range.

Interval Score (IS) — Winkler Score

$IS = \text{interval width} + (2/\alpha) \times \text{penalty for each missed observation}.$

This is the single best metric — it captures the coverage–width tradeoff in one number. A wide interval gets penalized for width; a narrow interval that misses gets heavily penalized for misses. Lower IS = better calibration.

The goal: narrowest possible intervals that still achieve the target coverage rate.

SECTION 7

Coverage Experiments & Per-Site Results

Coverage Results: All Sites × All Targets

Site	R ²	70% Target	80% Target	90% Target	95% Target
Bunchgrass Meadow	0.790	70.6%	80.7%	90.3%	94.6%
Easy Pass	0.545	69.7%	79.8%	89.7%	94.5%
Paradise	0.677	70.0%	79.2%	89.7%	94.6%
Touchet	0.893	71.3%	79.9%	89.8%	94.4%
Swift Creek	0.276	70.5%	79.9%	89.1%	93.8%

All sites achieve within 1–3% of every coverage target.

The 180-day rolling calibration window combined with asymmetric dual regulators ($\mu_2 \gg \mu_1$) consistently maintains coverage across diverse flow regimes. Even Swift Creek ($R^2 = 0.276$) achieves 93.8% at the 95% target — demonstrating that DACP works regardless of base model accuracy.

Average Interval Width Across Targets

Site	70% Width	80% Width	90% Width	95% Width	Units
Bunchgrass Meadow	490.46	596.99	742.66	816.43	cfs
Easy Pass	12.30	16.94	29.14	47.03	cfs
Paradise	4.71	6.36	10.06	14.86	cfs
Touchet	9.42	13.14	19.12	28.31	cfs
Swift Creek	2.36	3.46	5.96	9.78	cfs

Width increases monotonically with coverage target — higher coverage requires wider intervals.

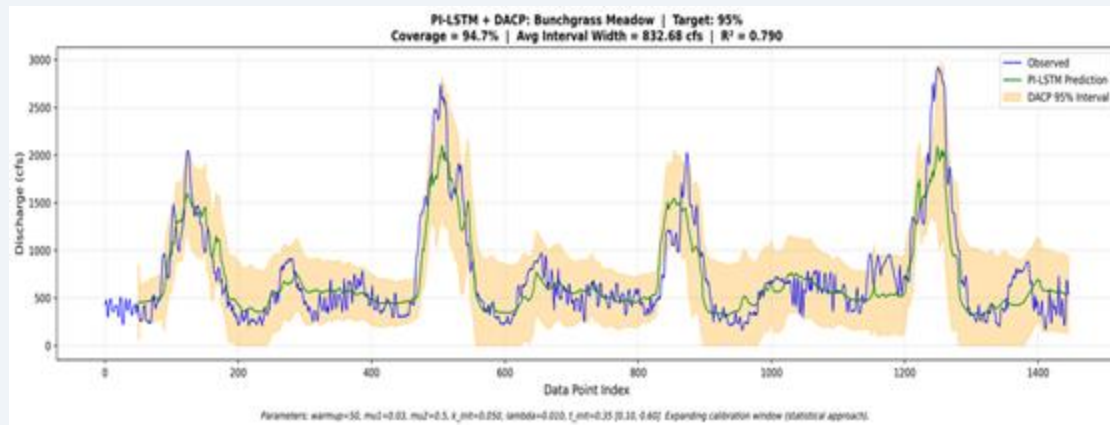
Bunchgrass's 816 cfs width at 95% is ~18% of its max discharge (4,500 cfs).

Swift Creek's 9.78 cfs is ~6% of its max (170 cfs).

Touchet's 28.31 cfs is only ~4.2% of max — the tightest relative width.

Relative widths are comparable across sites, showing DACP adapts to each site's scale.

Site Analysis: Bunchgrass Meadow

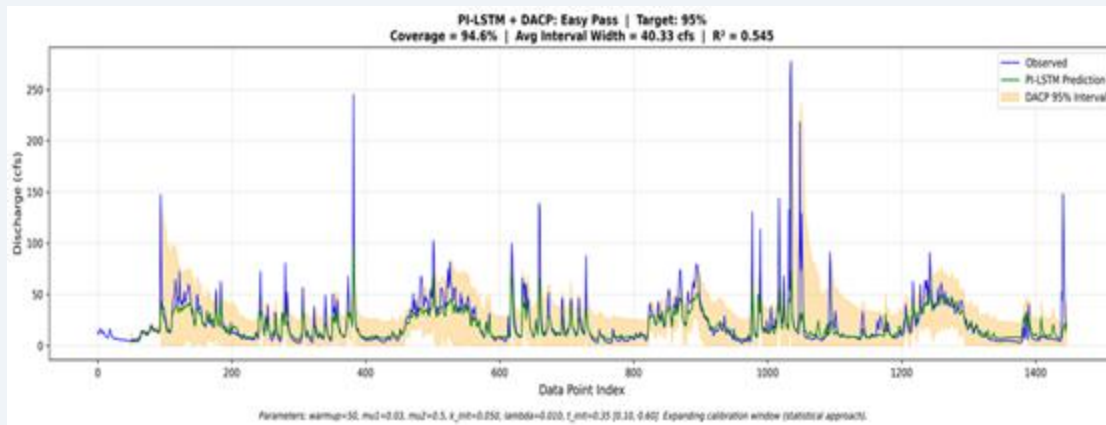


R^2 :	0.790
RMSE:	238.73 cfs
MRE:	0.380
Coverage:	94.6%
Avg Width:	816.43 cfs
Feature Set:	F2 (API)
Loss:	NSE
λ (Physics):	0 (disabled)

Highest-flow site with discharge up to 4,500 cfs. The model captures both spring snowmelt peaks and summer recession. RMSE of 238.73 cfs appears large but represents only ~5% of max discharge.

The $\log(1+Q)$ transform was essential to stabilize training on this extreme dynamic range. Physics loss was disabled ($\lambda = 0$) because in log space, the water balance term produces near-zero gradients — it becomes noise rather than signal. Average width of 816 cfs at 95% is proportional to the site's range (~18% of max).

Site Analysis: Easy Pass

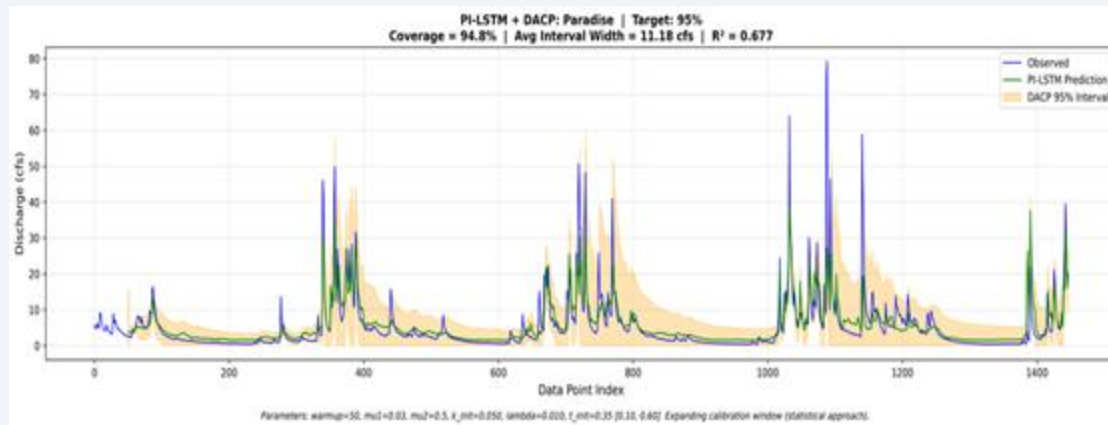


R^2 :	0.545
RMSE:	15.73 cfs
MRE:	0.492
Coverage:	94.5%
Avg Width:	47.03 cfs
Feature Set:	F1 (prcp_7d)
Loss:	Huber $\delta=1.0$
λ (Physics):	Site-specific

High-elevation alpine site where GHI (solar radiation) drives rapid snowmelt onset. $R^2 = 0.545$ is moderate — the model systematically underestimates during May–June because the rapid solar transition isn't fully captured by the 14-day window.

Huber loss was selected by Optuna to limit the influence of these peak underpredictions. Despite moderate R^2 , DACP achieves 94.5% coverage — the intervals correctly widen during high-uncertainty peak periods and narrow during stable baseflow. Average width of 47 cfs at 95% is ~17% of max (275 cfs).

Site Analysis: Paradise

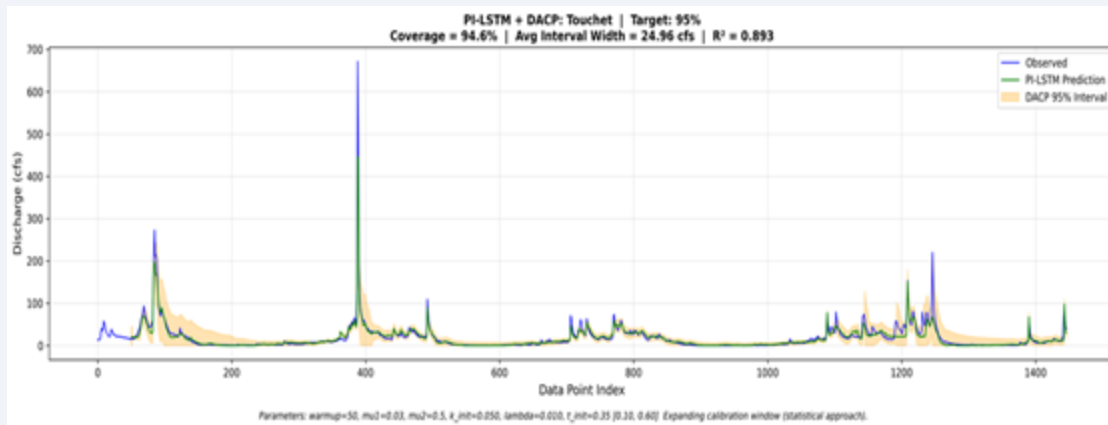


R^2 :	0.677
RMSE:	4.31 cfs
MRE:	0.886
Coverage:	94.6%
Avg Width:	14.86 cfs
Feature Set:	F1 (prcp_7d)
Loss:	Huber $\delta=1.0$
λ (Physics):	Site-specific

Rain-on-snow and rapid-response hydrology produces sharp, brief discharge spikes. $R^2 = 0.677$ is reasonable, but MRE = 0.886 is the highest — the model frequently mistimes individual flash events.

The 7-day precipitation feature captures rain-on-snow patterns better than API. RMSE of 4.31 cfs is small in absolute terms (site max ~ 35 cfs). High MRE despite decent R^2 shows the model captures overall variance well but makes large relative errors on low-flow days. DACP adapts quickly after misses ($\mu_2 = 0.50$).

Site Analysis: Touchet

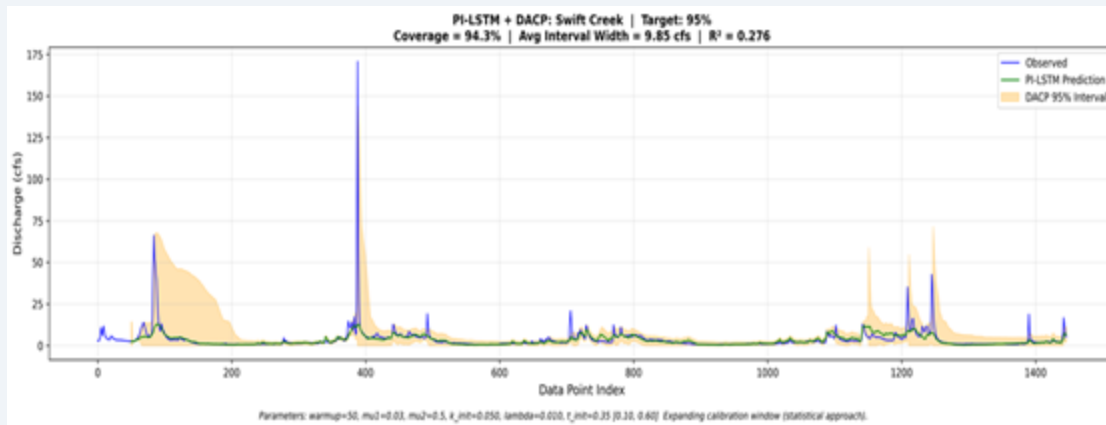


R^2 :	0.893
RMSE:	11.00 cfs
MRE:	0.505
Coverage:	94.4%
Avg Width:	28.31 cfs
Feature Set:	F1 (prcp_7d)
Loss:	Huber $\delta=1.0$
λ (Physics):	Site-specific

Best R^2 of all sites (0.893) — the model captures ~89% of daily variance. The dominant feature is a massive peak (~680 cfs) that dwarfs normal discharge (~20–50 cfs).

Huber loss was critical: it prevents the single extreme peak from dominating training. The DACP band correctly widens around the peak event and maintains tight intervals during stable low-flow periods. Average width of 28.31 cfs at 95% is only ~4.2% of max — tightest relative width of all sites.

Site Analysis: Swift Creek



R^2 :	0.276
RMSE:	5.93 cfs
MRE:	0.259
Coverage:	93.8%
Avg Width:	9.78 cfs
Feature Set:	F2 (API)
Loss:	MSE
λ (Physics):	0 (disabled)

Most challenging site — required fundamentally different treatment. Uses plain LSTM (no physics term, $\lambda = 0$) because the water balance constraint dominated the loss at low flows and degraded performance.

$R^2 = 0.276$ means only ~28% of variance is explained. However, $MRE = 0.259$ is the best of all sites. Key insight: DACP still achieves 93.8% coverage even with weak point predictions — the strongest demonstration that DACP provides reliable UQ regardless of base model accuracy.

Methodology Evolution: Key Decisions

Report 2 Static Split

Mean $R^2 = 0.756$. Bunchgrass $R^2 = 0.951$ — but optimistic due to temporal data leakage in the train/test split.

Report 3 Walk-Forward

Mean $R^2 = 0.671$. More realistic but harder. Fixed bugs: data leakage in scalers, random seed consistency, early stopping patience.

Report 4 Initial DACP

Used fixed rolling window with $\mu_1 \approx \mu_2$ from Optuna — achieved only 80–85% coverage at 95% target. Symmetric regulators couldn't react fast enough to misses.

Final Tuned DACP

180-day rolling window + asymmetric $\mu_2 \gg \mu_1 \rightarrow 94+\%$ coverage. With stable quantile estimation and active dual adaptation.

Deliverables

Complete Code Files

PIML_LSTM_v3.py (model training), dacc_retuned.py (DACP pipeline), run_all_v3.py (batch runner) — all fully commented.

Coverage Experiments

70%, 80%, 90%, 95% — all 20 experiments (5 sites $\times$ 4 targets) completed with updated α_m initial value per target.

Per-Site Plots & Tables

Hydrograph with DACP intervals for each site at each target. Tables showing coverage rate, average width, and model accuracy metrics.

CSV Data Files

Named PILSTM_SiteName_Target.csv with columns: true_value, prediction, upper_bound, lower_bound. Organized in per-site folders.

Final Report & Slides

Complete report document and this presentation, uploaded to Final Result.

Conclusion

- **PI-LSTM + DACP pipeline** developed for streamflow forecasting with uncertainty quantification across five snowmelt-dominated watersheds.
- **PI-LSTM with walk-forward prediction**, site-specific feature sets, and Optuna-tuned hyperparameters achieves R^2 from 0.276 (Swift Creek) to 0.893 (Touchet).
- **DACP with asymmetric dual regulators** ($\mu_2 \gg \mu_1$) and a 180-day rolling calibration window achieves 93.8–94.6% coverage at the 95% target across all five sites.
- **Coverage targets of 70%, 80%, 90%** are similarly met within 1–3%, demonstrating the DACP configuration generalizes across coverage levels.
- **DACP provides reliable UQ** even when point predictions are weak — Swift Creek ($R^2 = 0.276$) still achieves 93.8% coverage.

The 180-day rolling window provides stable quantile estimation while keeping the dual adaptive mechanism.

Future Considerations & Next Steps

1 Longer Input Sequences

Current 14-day window may miss slow snowpack accumulation over weeks. Extending to 30–60 days lets the model see longer moisture buildup, especially for glacial sites like Swift Creek. Trade-off: longer sequences need reduced model complexity to avoid overfitting on ~1,461 points per site.

2 Optuna Hyperparameter Tuning for DACP

Fixed parameters ($n=180$, $\mu_1=0.03$, $\mu_2=0.50$) work across all sites, but per-site optimization of n and adaptive speeds could tighten intervals. Initial Optuna tuning failed with symmetric μ values — a constrained search enforcing $\mu_2 \gg \mu_1$ and site-specific n could improve coverage-width trade-offs.

3 Temporal Fusion Transformer (TFT)

TFT has multi-horizon attention — it learns that yesterday's temperature matters differently than last week's precipitation. Better at long-range dependencies than LSTM. Challenge: transformers are data-hungry with only ~1,461 days per site. Feasible via multi-site pretraining (shared physics, different scales), then per-site fine-tuning.

All three directions build on the PI-LSTM + DACP framework while addressing current limitations in sequence length, parameter tuning, and architecture.

Thank You

PI-LSTM + DACP: Streamflow Prediction
with Uncertainty Quantification